Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: **Prod Pilots**
b. Team leader name: **Lohith DS**
c. Problem Statement: BFSI: Intelligent Risk, Fraud and Financial Experiences. Rakshak, a scam-moment guardian that stops authorised-push-payment scams at the moment of payment.

Brief about the idea

Rakshak: stop the scam at the moment of payment, not after it

Scam victims now send the money themselves, on a call, under pressure. Bank fraud engines see a genuine customer on a genuine device, and an SMS alert arrives after the money has left.

Rakshak is a guardian built into a bank or UPI app. With the customer's opt-in it listens to a call from an unknown number and scores the caller's coercion live with Gemini. When the customer taps Pay, it fuses that call risk with the payee's mule-account risk from the bank's ledger, and in under 100 ms pauses the payment, explains why in the customer's language, alerts a trusted contact and files an auditable case. An ADK agent confirms with evidence; the customer always keeps an override.

Built and deployed

- Customer app (installable PWA), caller phone, family phone and fraud desk on Cloud Run
- Gemini 3.5 and 3.8 models, Google ADK 2.11 agents, Firestore case store
- Live two-way calls between two phones, six languages, 50-call evaluation

81.8%

of Singapore's 2025 scam cases were transfers the victim made (Singapore Police)

₹22,495 cr

lost to cyber fraud in India in 2025, about 24% recovered (MHA via Moneylife)

₹1.56 lakh

average loss per digital-arrest victim in 2024 (Lok Sabha data)

Opportunities

How is it different from existing ideas?

- Pixel scam detection: on the handset, warns only, knows nothing about the payee
- Call-state signals (FICO Scam Signal, BioCatch): know a call is on, not what is said
- Bank fraud engines: blind to the call; the coerced victim passes every check
- SMS alerts and 1930: after the money left; about 24% is recovered
- Rakshak fuses call semantics with payee mule risk at the moment of authorisation, on the bank side

How does it solve the problem?

- Listens (opt-in) from the first ring; scores 8 coercion signals live in 6 languages
- Whispers privately to the customer when the risk passes 85%
- At Pay: graded decision in under 100 ms: proceed, warn, 30-minute hold, or hold plus alert
- Speaks the reason, notifies a trusted contact, drafts the fraud-desk case and 1930 referral
- Payments within 60 minutes of a high-risk call are still checked

USP of the solution

- Intervention, not an alert: the money does not leave until the customer has had time to think
- Explainable: every hold names its evidence; an ADK agent can only make it stricter
- Human in the loop: the customer can override by typing the amount
- Vernacular voice for the people scammers target most
- Privacy tiers: works with no audio, on-device text, or cloud analysis

List of features offered by the solution

1 Live coercion scoring

Gemini scores 8 signals (authority, threat, secrecy, urgency, remote app, safe account, investment pressure, OTP) after every sentence

2 Private whisper

At 85% risk Rakshak warns the customer in their ear; the caller hears nothing

3 Payment-moment hold

Deterministic fusion of call risk, payee risk and profile decides in under 100 ms

4 Voice barge-in

The reason is spoken in the customer's language; Hindi, Hinglish, English, Kannada, Tamil, Singapore English

5 Payee mule risk with lineage

Fan-in, cash-out, dormancy, account age, telecom risk and 1930 complaints, computed from a live ledger stream

6 ADK investigator

Two Gemini agents check the payee and official advisories (RBI, CBI, MAS, SEBI) and confirm or upgrade

7 Trusted-contact alert

The family phone chimes with the case; also Pub/Sub and SMS hooks

8 Talk to Rakshak

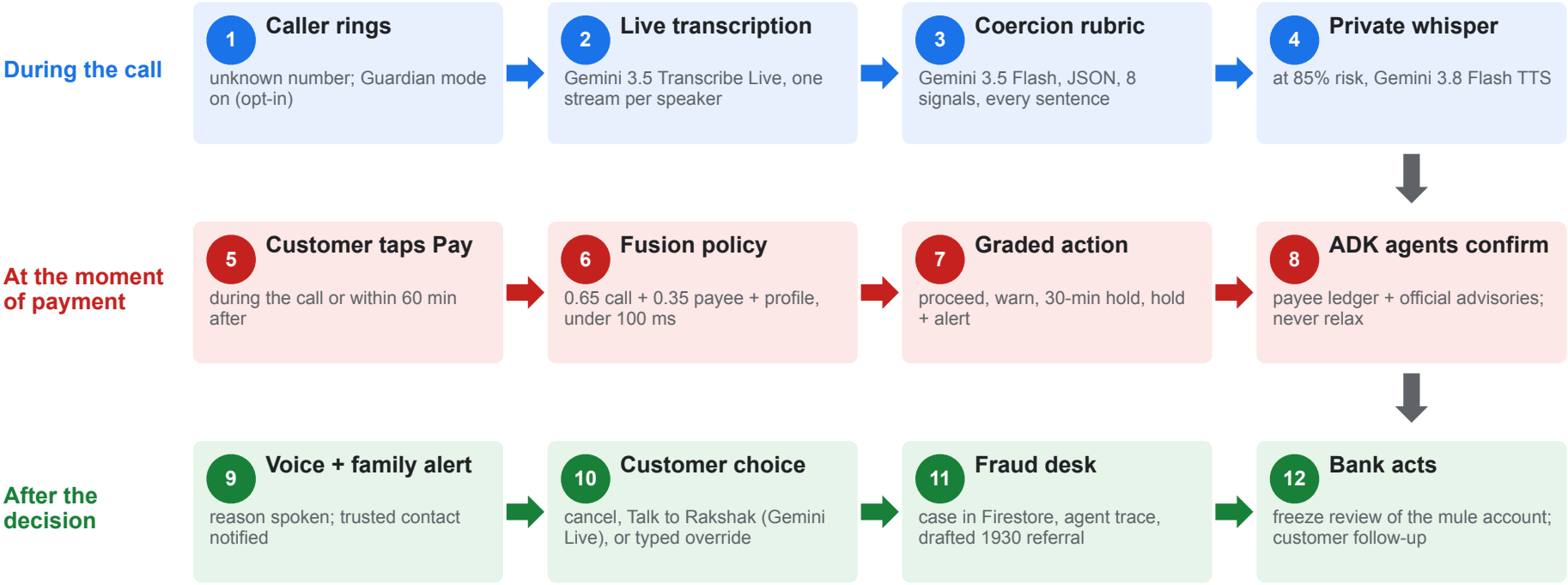
Gemini Live voice conversation after the hold, with tools to keep the hold or call family

9 Fraud desk

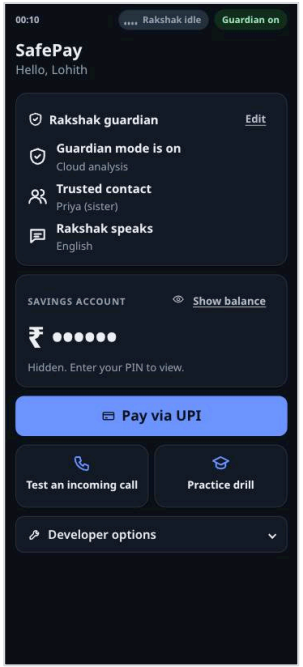
Case timeline, agent trace, drafted 1930 referral, and the SMS-versus-Rakshak contrast

Also: post-call carry-over (60 minutes), practice drill for elderly users, PIN-hidden balance, live two-way call between phones, installable app.

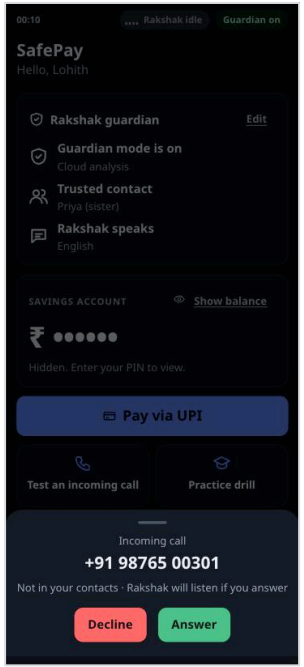
Process flow diagram or Use-case diagram



Wireframes/Mock diagrams of the proposed solution (optional)



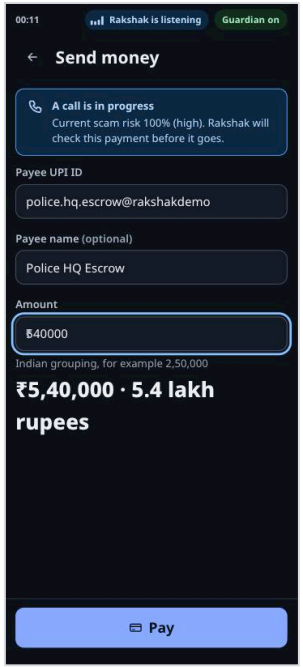
Home: Guardian on, balance hidden by PIN



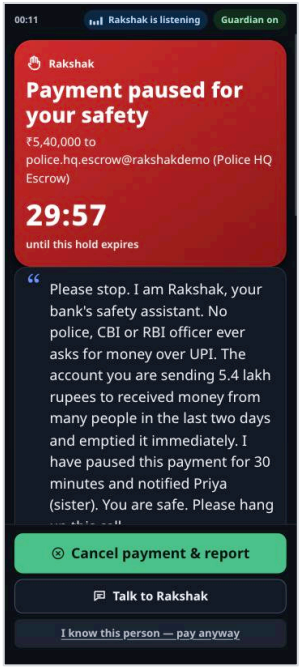
Unknown number rings



Live risk 100%, whisper, reasons

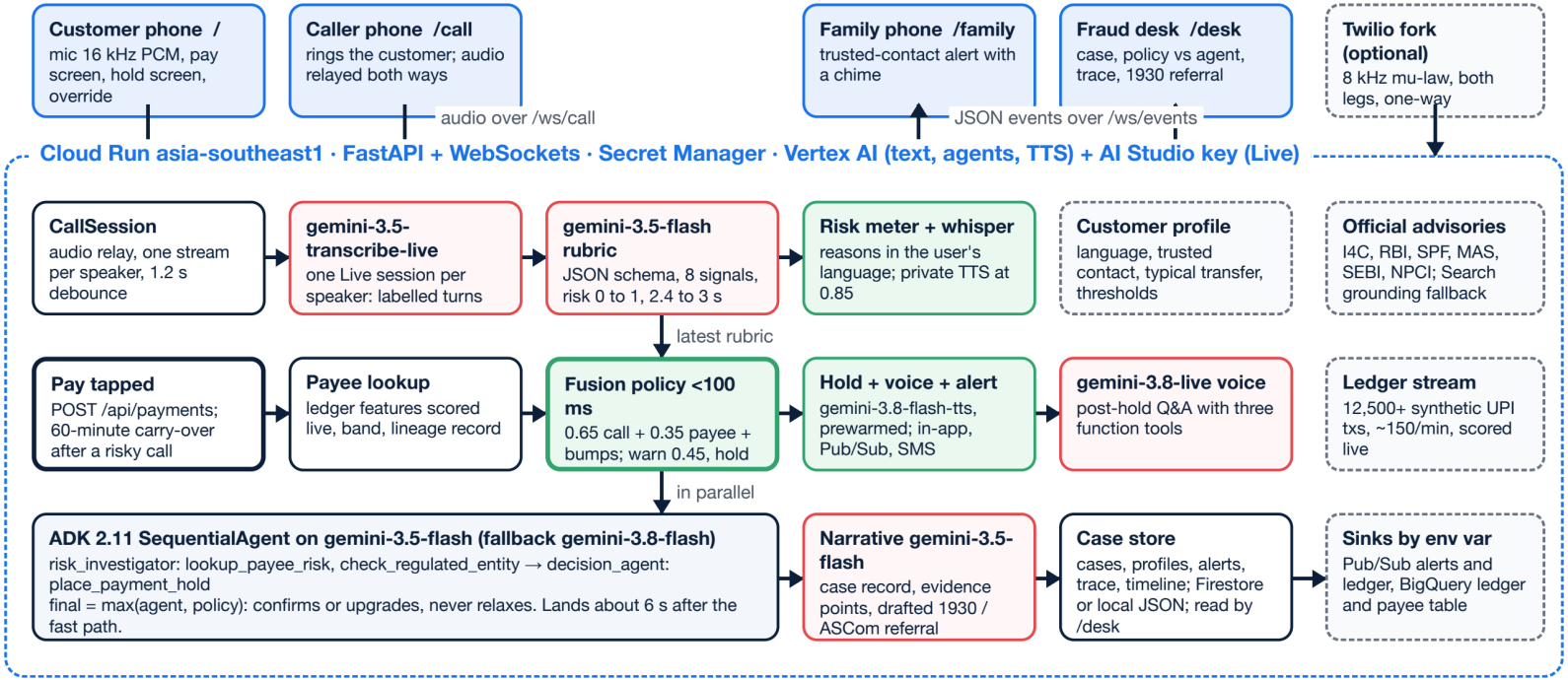


Pay screen knows a call is on



Payment paused, reason spoken

Architecture diagram of the proposed solution



Google Cloud: Cloud Run (FastAPI, WebSockets, one always-on instance) · Gemini on Vertex AI and the Gemini API · ADK 2.11 · Firestore case store (Firebase) · Pub/Sub alerts · BigQuery ledger SQL · Secret Manager. Firebase Hosting serves the same app at https://fde-cloud-run-lab-123456.web.app.

Technologies to be used in the solution

Gemini models

- 3.5 Flash: coercion rubric (JSON), ADK agents, case narrative
- 3.5 Transcribe Live: one live stream per speaker
- 3.8 Flash TTS: whisper and spoken hold
- 3.8 Live: voice conversation with tools
- 3.8 Flash: agent fallback

Agents

- Google ADK 2.11 SequentialAgent
- risk_investigator: payee ledger + official advisories, Search grounding fallback
- decision_agent: place_payment_hold tool
- Policy floor: agents can only make it stricter

Serving

- Cloud Run, asia-southeast1, WebSockets, session affinity
- FastAPI, Python 3.12, uv
- Installable PWA: customer app, caller phone, family phone, fraud desk
- Firebase Hosting front door

Data

- Firestore (Firebase): cases, profiles, alerts
- Ledger stream with mule features (same formula as the BigQuery SQL)
- Pub/Sub for trusted-contact alerts; BigQuery sink optional

Security and privacy

- Secret Manager for the API key; IAM service account
- Opt-in Guardian mode, visible indicator
- No raw audio stored; PIN-hidden balance
- Typed-amount override, holds expire

Quality

- 50-call labelled evaluation in 6 languages
- 16 automated tests incl. a 3-day ledger drift test
- Live two-device audio smoke test on the public URL

Estimated implementation cost (optional)

Item (per guarded 3-minute call)	Assumption	Estimate (USD)
Live transcription, both speakers	Gemini 3.5 Transcribe Live, 360 s of audio at 25 tokens/s	0.03
Coercion rubric	about 15 scorings, Gemini 3.5 Flash	0.07
ADK agents and case narrative	about 5 Gemini 3.5 Flash calls	0.03
Whisper and spoken hold	Gemini 3.8 Flash TTS, about 33 s of audio	0.01
Total per guarded call with a hold	list prices, Gemini API, Oct 2026	about 0.14 (₹12)

Infrastructure (prototype)	Estimate
Cloud Run, 1 always-on instance (1 vCPU, 1 GiB)	about USD 50 to 70 a month
Firestore, Pub/Sub, BigQuery, Secret Manager	within free tiers at prototype scale

₹12

per guarded call that ends in a hold

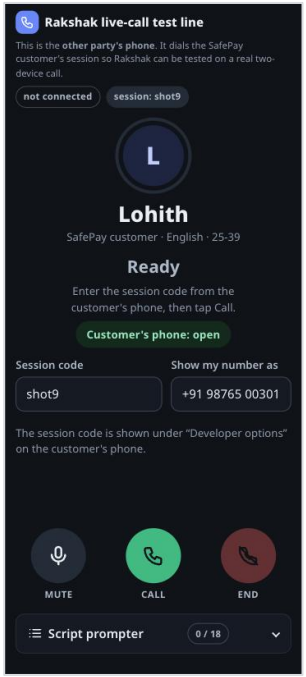
versus ₹1.56 lakh lost by the average digital-arrest victim

Cost is paid only when Guardian mode is on and an unknown number calls.

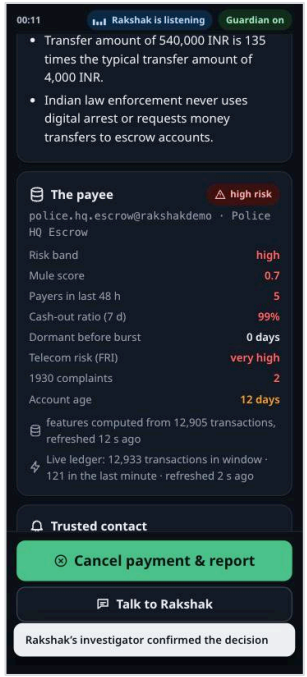
At bank scale

- Sessions move to Firestore or Memorystore and Cloud Run scales out; agents can move to Agent Runtime.
- Estimates use list prices; Vertex AI contract prices differ.

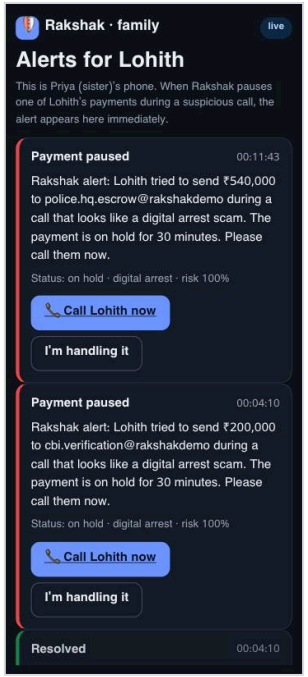
Snapshots of the prototype



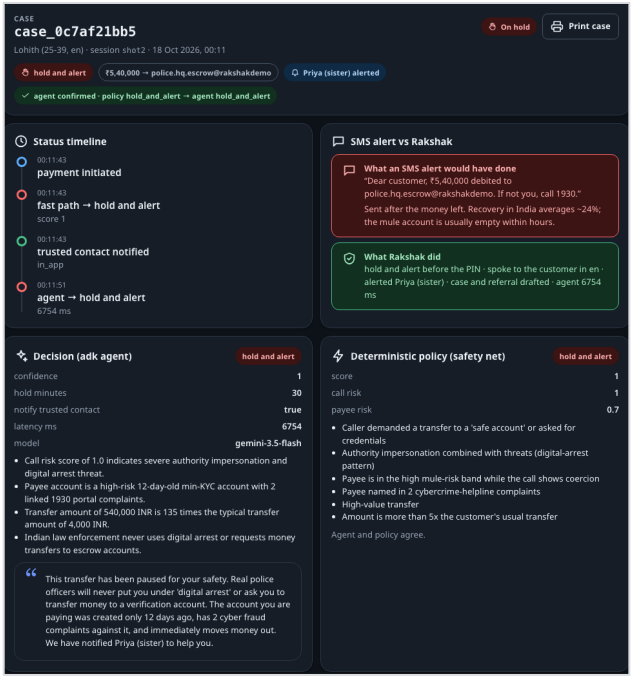
Caller phone: live two-way call



Payee evidence with ledger lineage



Family phone: alert arrives



Fraud desk: case, timeline, agent versus policy, SMS contrast

Prototype Performance report/Benchmarking

1.00
Precision
no benign call held

1.00
Recall
every scam call held

39 / 39
Payment decisions
correct action on every payment

93.2%
Signal accuracy
8 signals x 50 calls

50 labelled calls	Predicted scam	Predicted benign
Actual scam (25)	25	0
Actual benign (25)	0	25

25 scam and 25 benign calls in Hindi, Hinglish, English, Singapore English, Kannada and Tamil, including 10 legitimate-but-urgent calls and 10 scams written to dodge keywords. Self-authored set, not held-out.

Latency and reliability	Measured
Coercion score after a sentence	3.1 s mean, 2.0 s median
Hold on screen after Pay	0.02 to 1.4 s
Spoken reason (prewarmed)	about 0.1 s after the hold
ADK agent confirmation	5.4 to 7.2 s
Live two-device audio test	3 of 3 sentences transcribed; whisper fired
Automated tests	16 passing, incl. 3-day ledger drift
Ledger stream	about 12,800 transactions in window, about 120 a minute

Additional Details/Future Development (if any)

Next with a bank or PSP partner

- Real ledger features in BigQuery on a schedule; DoT FRI, NPCI DPIP and I4C registry feeds
- On-device Tier 1 with an OS or OEM partner (Pixel scam detection to a GPay hold)
- Telco audio fork for cellular calls, as banks already do with call-state signals
- Native Android app: call-state signal, foreground service, FCM push to family
- Human-labelled call corpus and red-team evaluation; native-speaker QA for Indic voices
- Model-risk evidence pack for RBI FREE-AI and MAS AI risk guidelines

Honest limits of this prototype

- Synthetic ledger and scripted calls; no bank partner yet
- Third-party apps cannot tap cellular calls on Android or iOS, so live audio uses in-app calls or a telco bridge
- The Twilio phone bridge is one-way: the spoken hold stays on the app
- Kannada and Tamil voices not yet tested with native speakers
- One Cloud Run instance holds live call sessions; scale-out needs shared session state

Provide links to your:

- GitHub Public Repository: https://github.com/devaramanelohith/rakshak-agent
- Demo Video Link (3 Minutes)**: [paste the public YouTube link here]**
- Final Product Link: https://rakshak-223197895518.asia-southeast1.run.app

Try it in 60 seconds (one device)

- Open the product link, tap Test an incoming call
- Choose Cyber Crime Branch (English), tap Auto-play script
- When asked to pay, tap Open UPI pay, then Pay
- Watch the hold, hear Rakshak, open /desk for the case
- Balance PIN for the demo: 1234

Live two-way call (two phones)

- Customer phone: /?session=demo1&user=demo-user
- Caller phone: /call?session=demo1, tap Call
- Family phone: /family?user=demo-user
- Fraud desk: /desk
- Hindi customer: &user=demo-user-hi

Privacy by design, and how it works on a real cellular call

Tier	What Rakshak uses	What leaves the phone
0: no audio	Call from an unknown number + pay screen + payee risk	Only the payment request
1: on-device text	Rolling 60 s transcript kept on the phone	Scores and last 60 s, only if a payment is attempted
2: cloud (this prototype)	Live transcription through a lawful channel	Audio stream, never stored

Rules at every tier

- Guardian mode is opt-in with a visible listening indicator
- The caller never hears Rakshak; holds expire; the customer can override
- Raw audio is never written to disk; cases are kept for the bank's audit trail

Real cellular calls

No third-party app can read cellular call audio on Android 10+ or iOS. Rakshak therefore runs on the bank side and receives audio only through lawful channels:

- the bank's own in-app or verified calls (shown in the demo)
- a telco audio fork with consent (FICO Scam Signal precedent)
- an OS vendor feed: Google can connect Pixel scam detection to a GPay hold

Without audio, Tier 0 still holds risky payments to mule accounts during unknown calls.

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Thank you

